Ex.No:14
08.10.26

14. WRITE CODE USING RAW SOCKETS TO IMPLEMENT PACKET SNIFFING

Aim:

To implement a packet sniffing program using raw sockets to capture network packets and display the protocol, source IP address, and destination IP address.

Packet sniffing using scapy:

```
from scapy.all import sniff
from scapy.layers.inet import IP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown protocol"
        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("")
sniff(
    iface = "Wi-Fi",
```

```
        prn = packet_callback,
        filter = "ip",
        store = 0
    )
```

O/P:

Protocol : TCP
Source IP : 192.168.1.5
Destination IP : 142.250.195.14

Protocol : UDP
Source IP : 192.168.1.5
Destination IP : 8.8.8.8

Protocol : ICMP
Source IP : 192.168.1.5
Destination IP : 8.8.8.8

Result:

Thus, the packet sniffing program was successfully implemented and the captured packets were analyzed.